

TECHNOLOGY SPECIFICATIONS DOCUMENT

Section 1: Large Language Models

GPT-4 was released in March 2023 by OpenAI. It uses a transformer architecture and has an estimated 1.7 trillion parameters, making it one of the largest language models ever created.

Section 2: Quantum Computing

IBM's Condor processor achieved 1,121 qubits in December 2023. Google's Sycamore processor has 53 qubits and demonstrated quantum supremacy in 2019.

Section 3: GPU Specifications

NVIDIA H100 GPU has 80GB of HBM3 memory and delivers 3,958 TFLOPS of FP8 performance. The H100 uses TSMC's 4nm process node.

Section 4: Battery Technology

Lithium-ion batteries have an energy density of approximately 250-300 Wh/kg. Solid-state batteries promise energy densities exceeding 400 Wh/kg.

Section 5: Internet Infrastructure

5G networks offer peak speeds up to 20 Gbps with latency as low as 1 millisecond. Starlink satellites orbit at approximately 550 km altitude.

Section 6: Semiconductor Manufacturing

TSMC produces chips at the 3nm process node as of 2024. EUV lithography uses 13.5nm wavelength light for chip manufacturing.